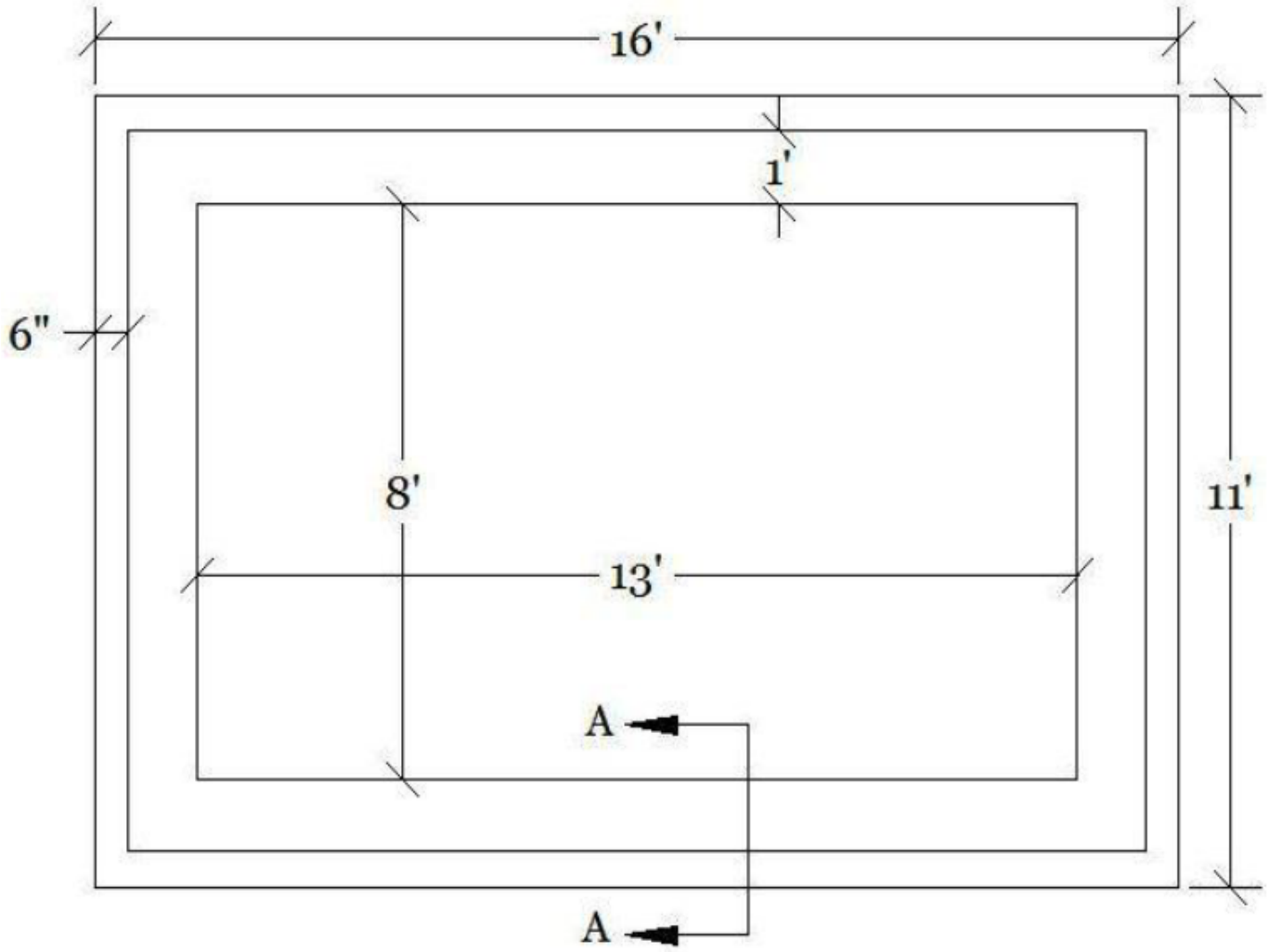


CE 208
Quantity Surveying

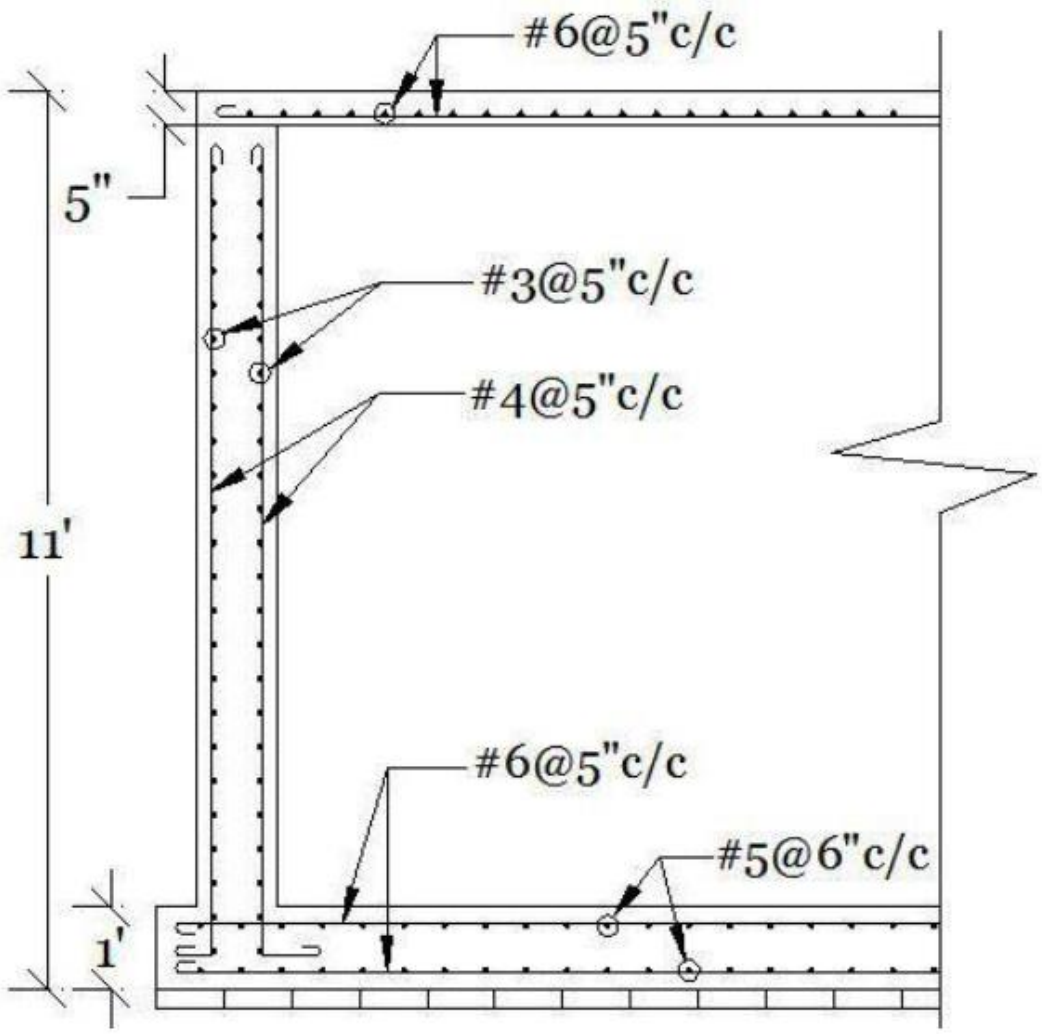
Estimation of an Underground Water Reservoir

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Plan View



Section A-A



Segments



- BFS (3”)
- Base Slab
- Wall
- Cover Slab

BFS (3″)

1. Length= 16′
2. Width= 11′

No. of Bricks

Size of one brick= 10″x5″x3″

Area of one brick= 50 in2

No. of brick required= 176x144/50

= 507 Nos.

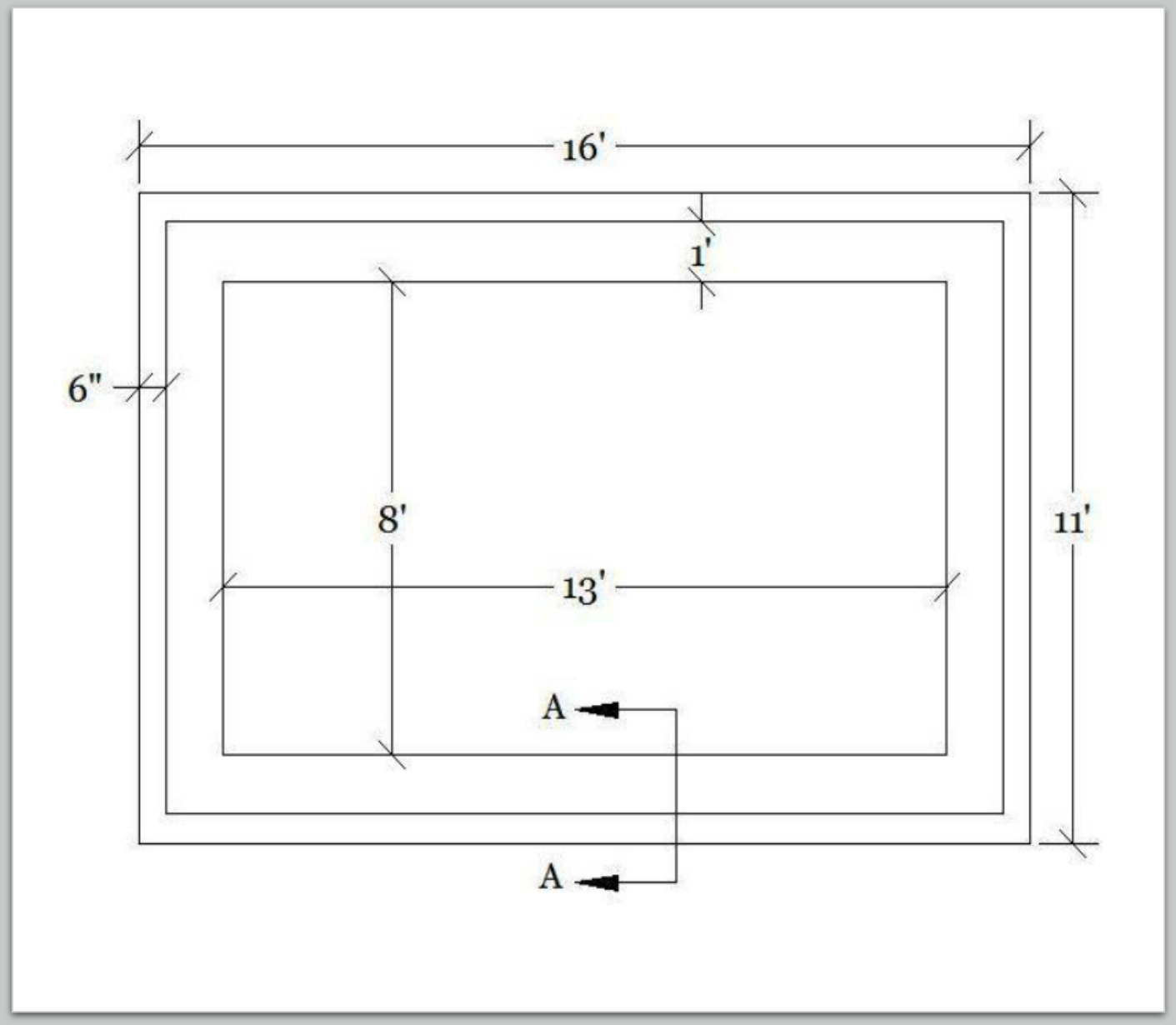
Volume of Sand fill

Sand per 10m2 = 0.10 m3

Volume of sand= [0.10x176x(0.3048)^2]/10

= 0.1635 m3

= 5.77 ft3



Base Slab

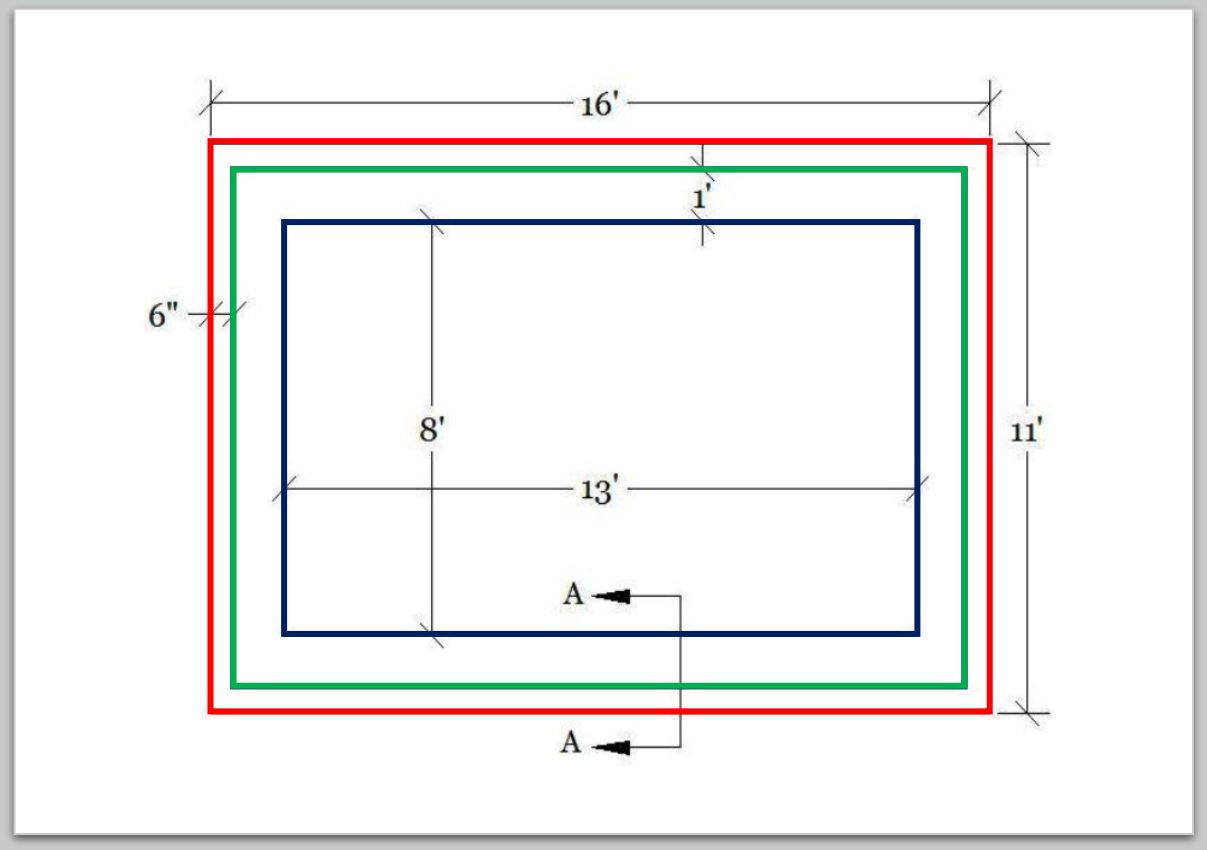
- Length= 16’
- Width= 11’
- Height= 1’
- Volume= 176 ft3

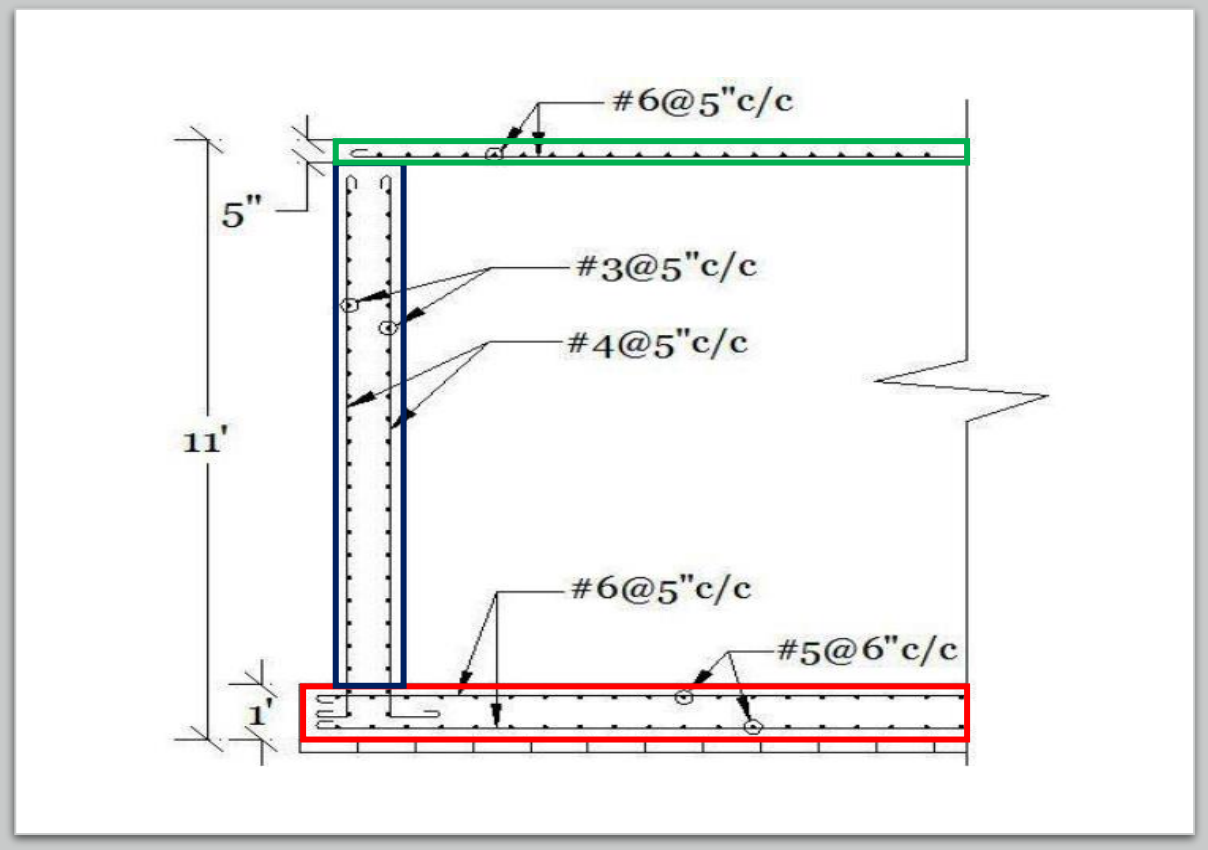
Wall

- Length= 2x(9’+14’)= 46’
- Width= 1’
- Height= 10’ -5”=9’7”
- Volume= 440.83 ft3

Cover Slab

- Length= 15’
- Width= 10’
- Height= 5”
- Volume= 62.5 ft3





Concrete (1:2:4)



1. Final volume (hard concrete) = 176 + 440.83 + 62.5= 679.33 ft3
2. Wet volume = 679.33 x1.5= 1018.995 ft3
3. Cement= (1018.995 x 1)/7= 145.57 ft3 = 117 bags
4. Sand= (1018.995 x 2) /7= 291.14 ft3
5. Brick chips= (1018.995 x 4)/7= 582.28 ft3= 4947 Nos. brick

[1m3= 35.315 ft3 brick chips required 300 Nos. of full-size brick]

Summary

1

Cement

117 bags

2

Sand

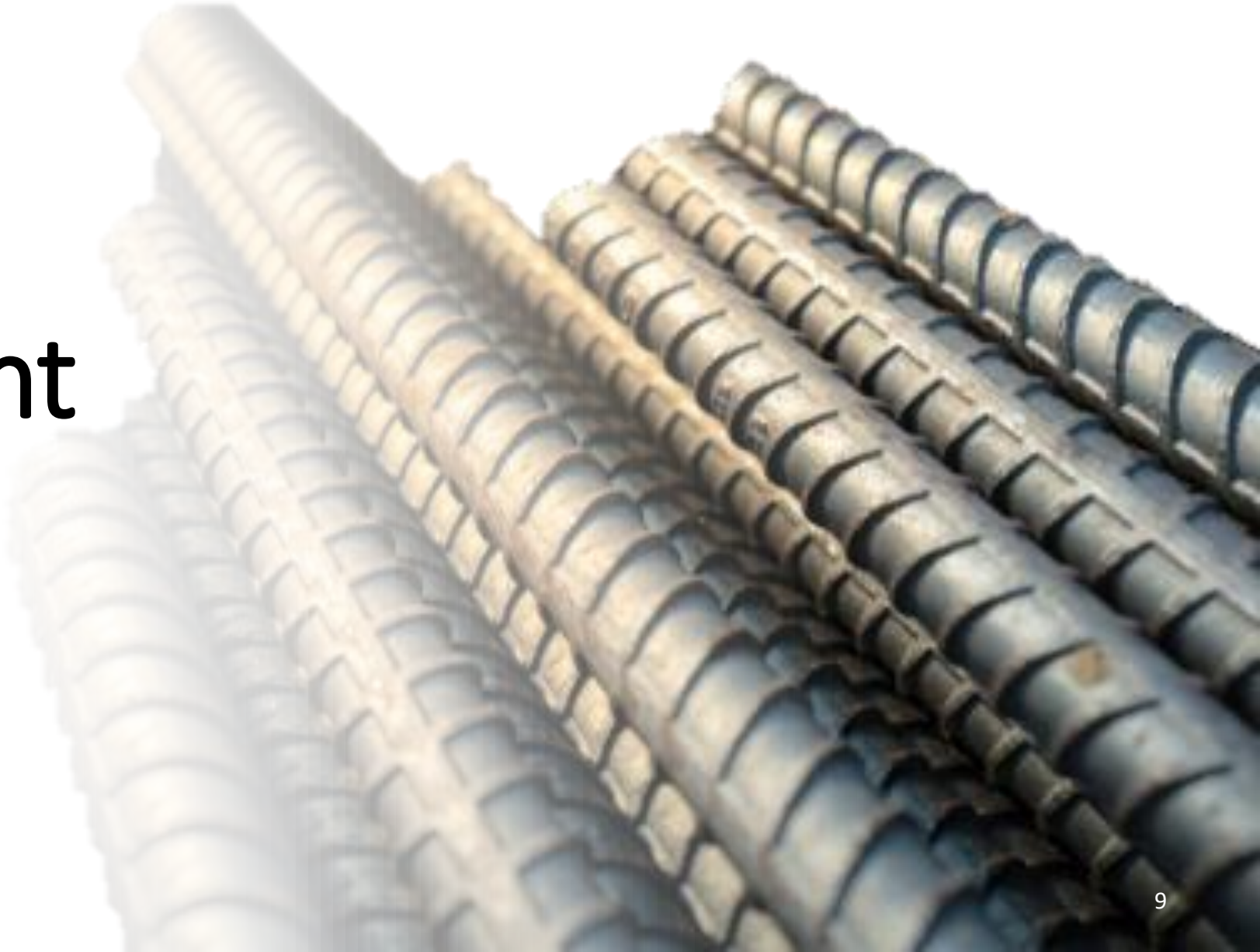
297 ft3

3

Brick

5314 Nos.

Reinforcement



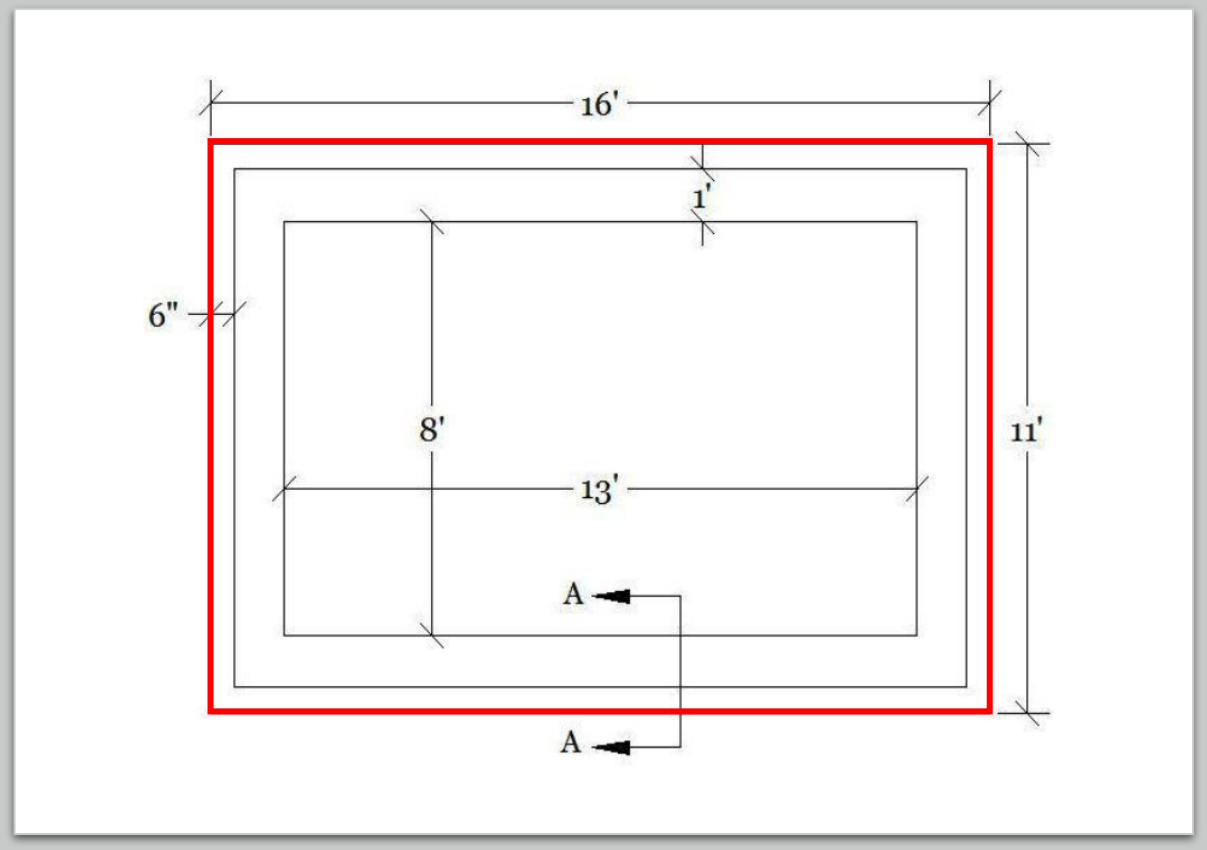
Base Slab

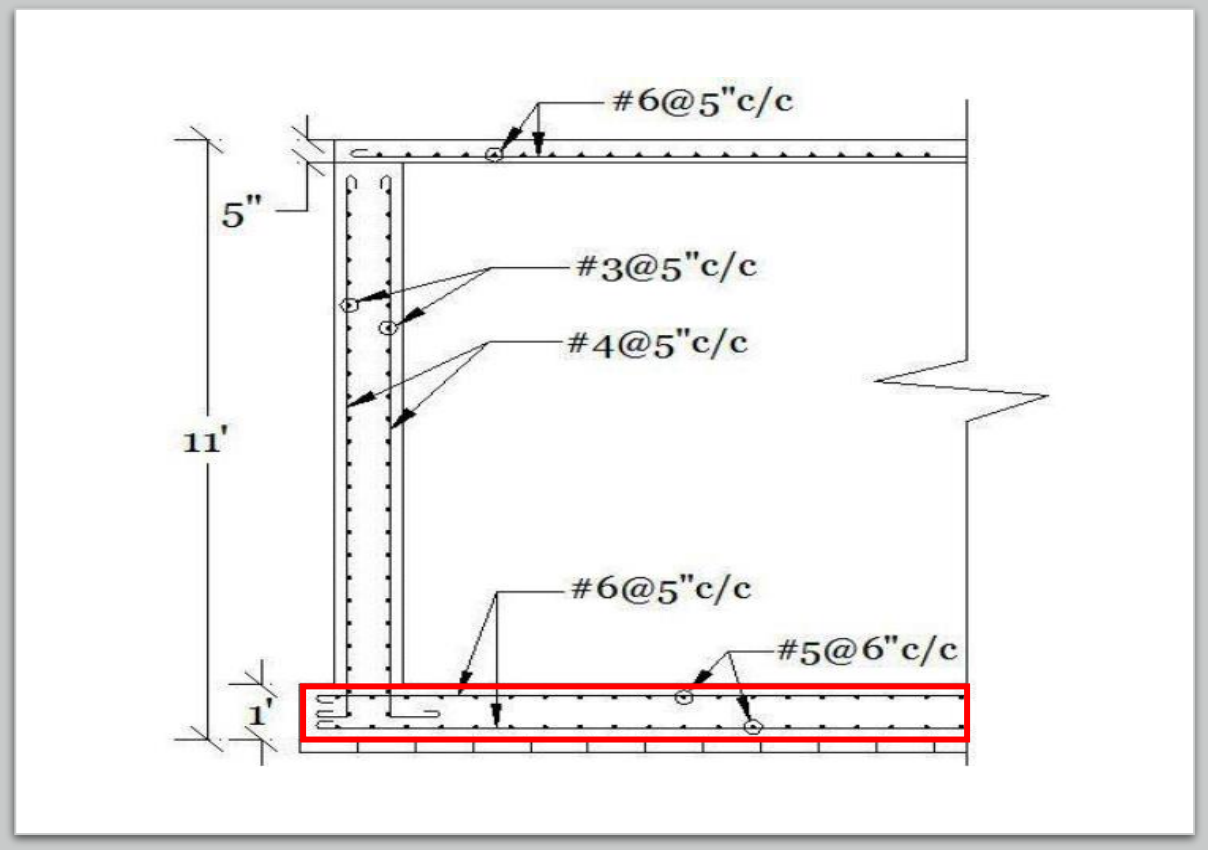
#6 @5”c/c

- No.= 2x[(16x12-2x3)/5+1]= 77
- Length= (11x12-2x3+2x10.5x6/8)/12= 11.813ft.
- Total length= 909.601ft.

#5@6”c/c

- No.= 2x[(11x12-2x3)/6+1]= 44
- Length= (16x12-2x3+2x10.5x5/8)/12= 16.594ft.
- Total length= 730.136ft.





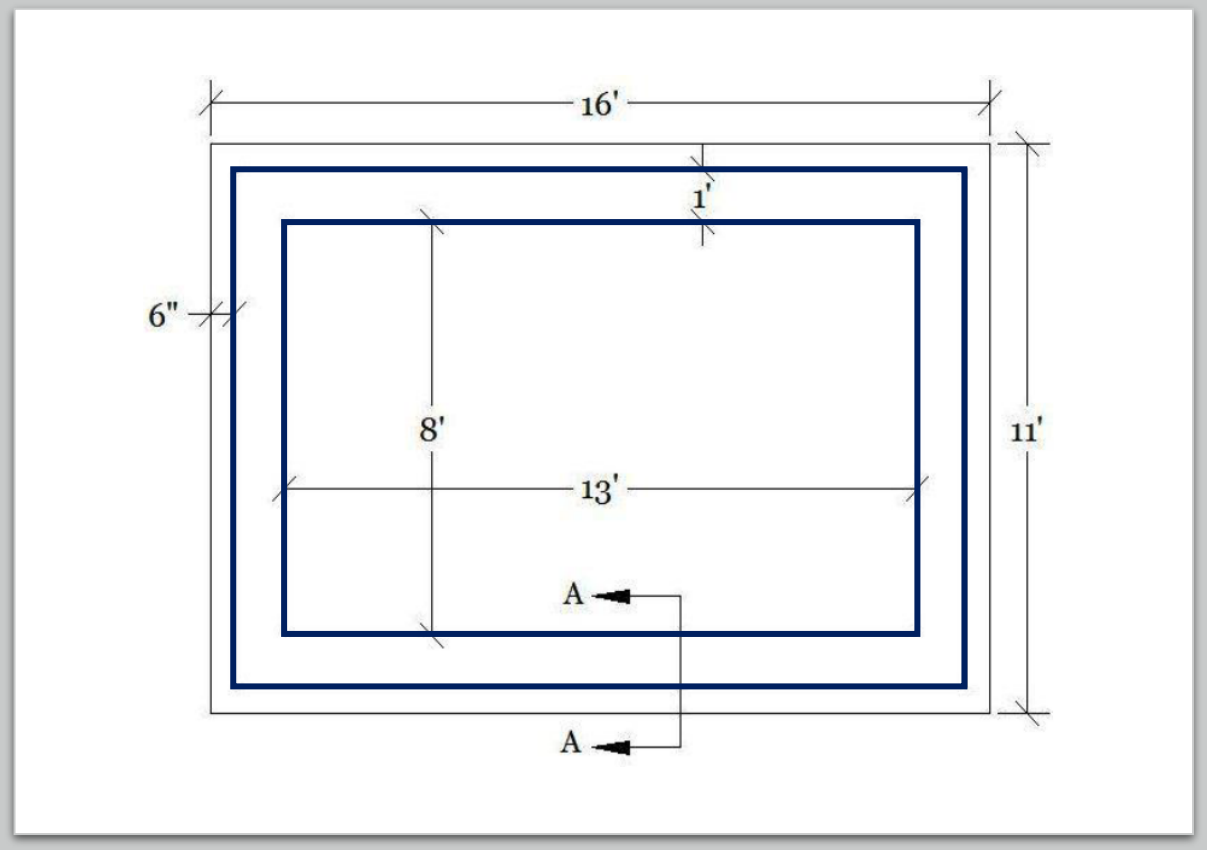
Wall

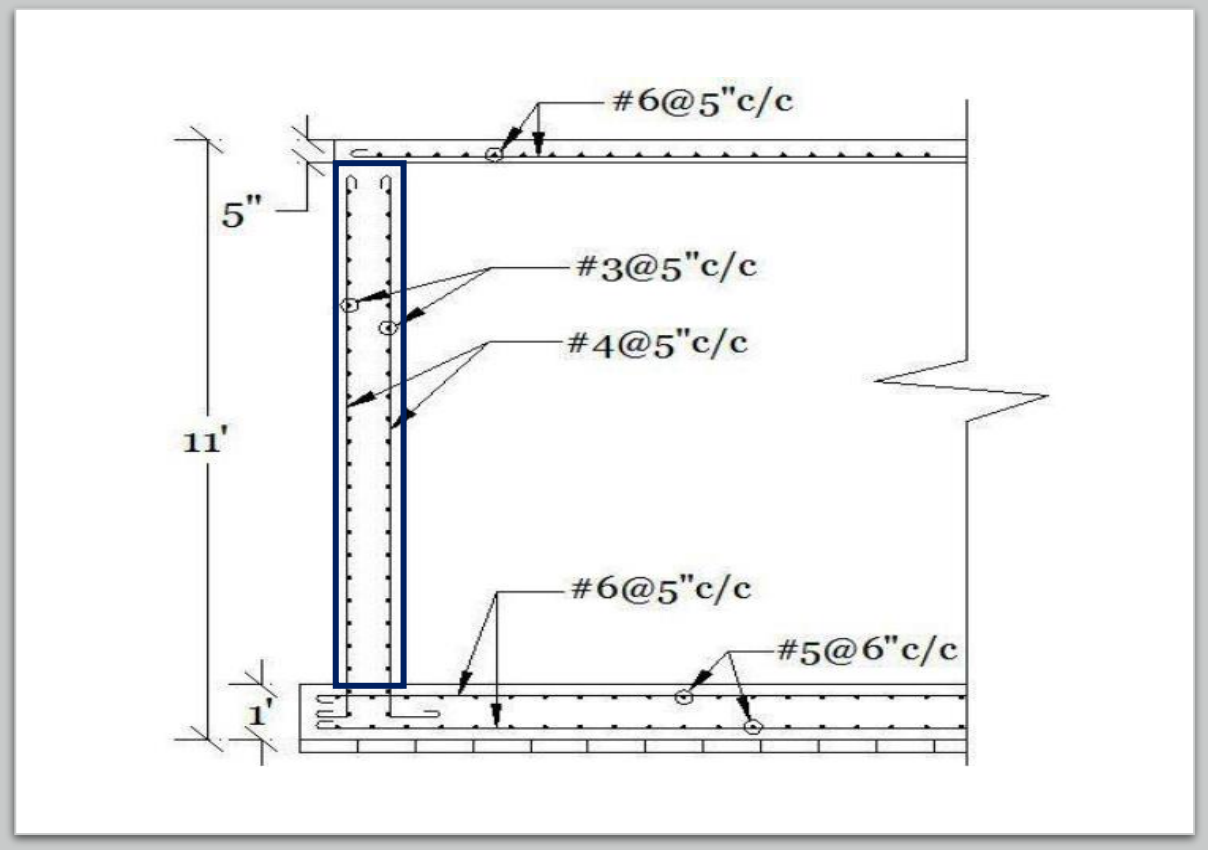
#4@5”c/c (outside)

- No.= 2x[(15x12/5+1)+(10x12/5-1)] = 120
- Length= (11x12-5-2-3+4+2x10.5x4/8)/12
  = 11.375ft.
- Total length= 1365ft.

#4@5”c/c (inside)

- No.= 2x[(13x12/5+1)+(8x12/5-1)] = 108
- Length= (11x12-5-2-3+6+2x10.5x4/8)/12
  = 11.542ft.
- Total length= 1246.536ft.

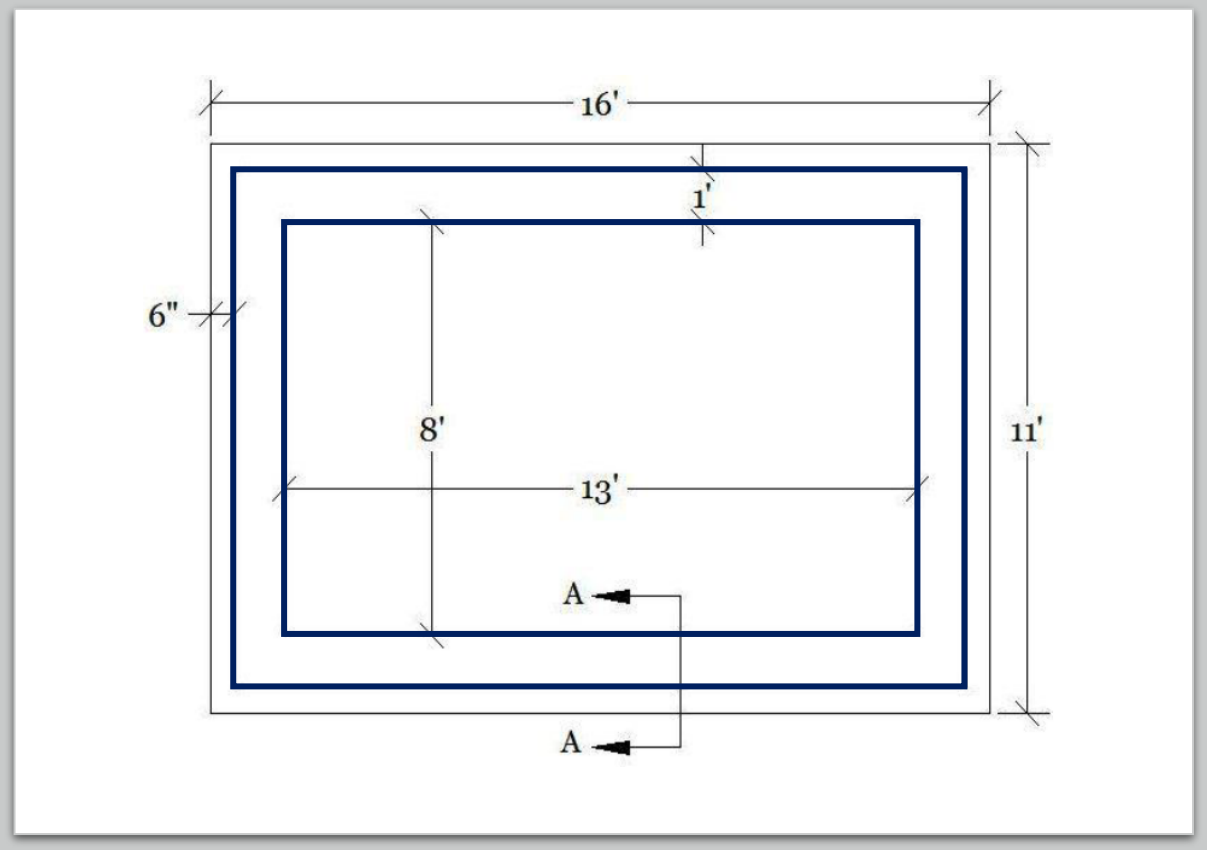


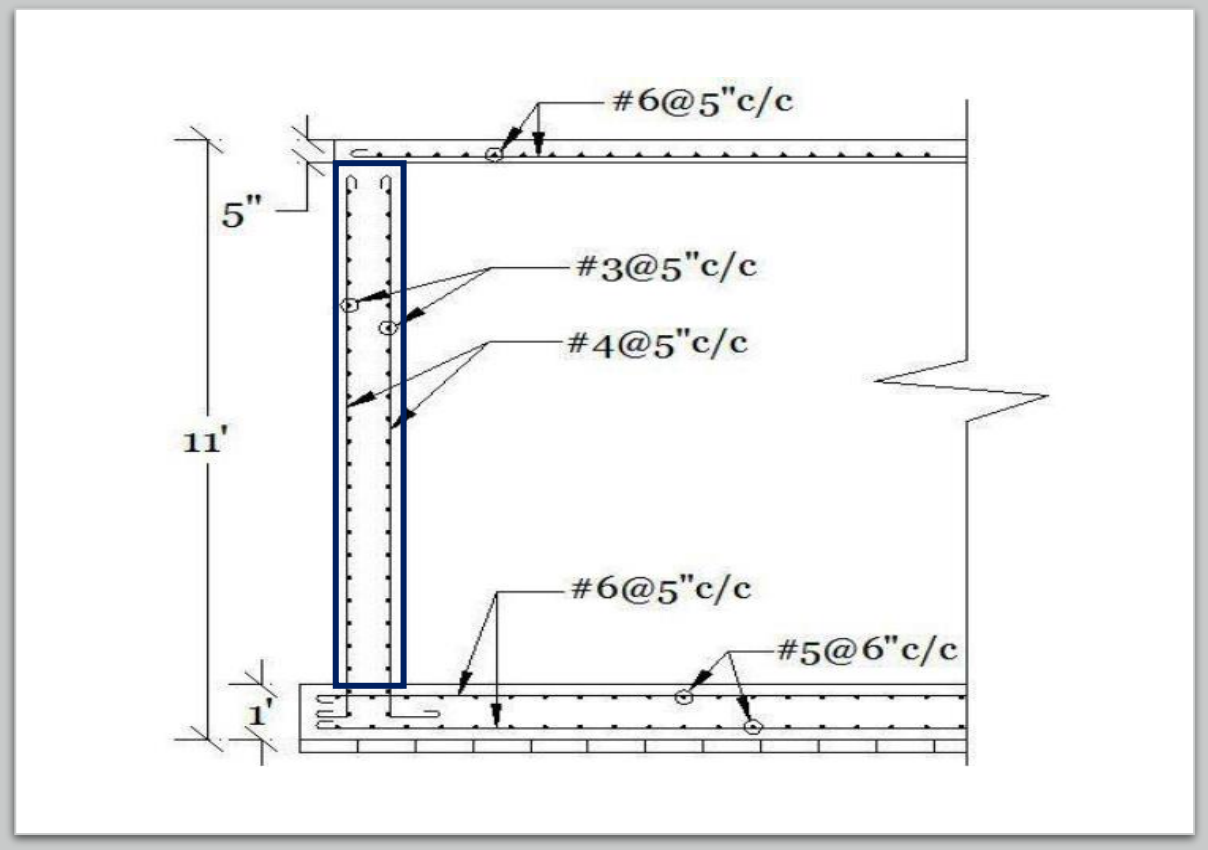


Wall

#3@5”c/c

- No.= 2x[(11x12-5-2-3)/5+1]= 52
- Length= 46ft.
- Total length= 2392ft.





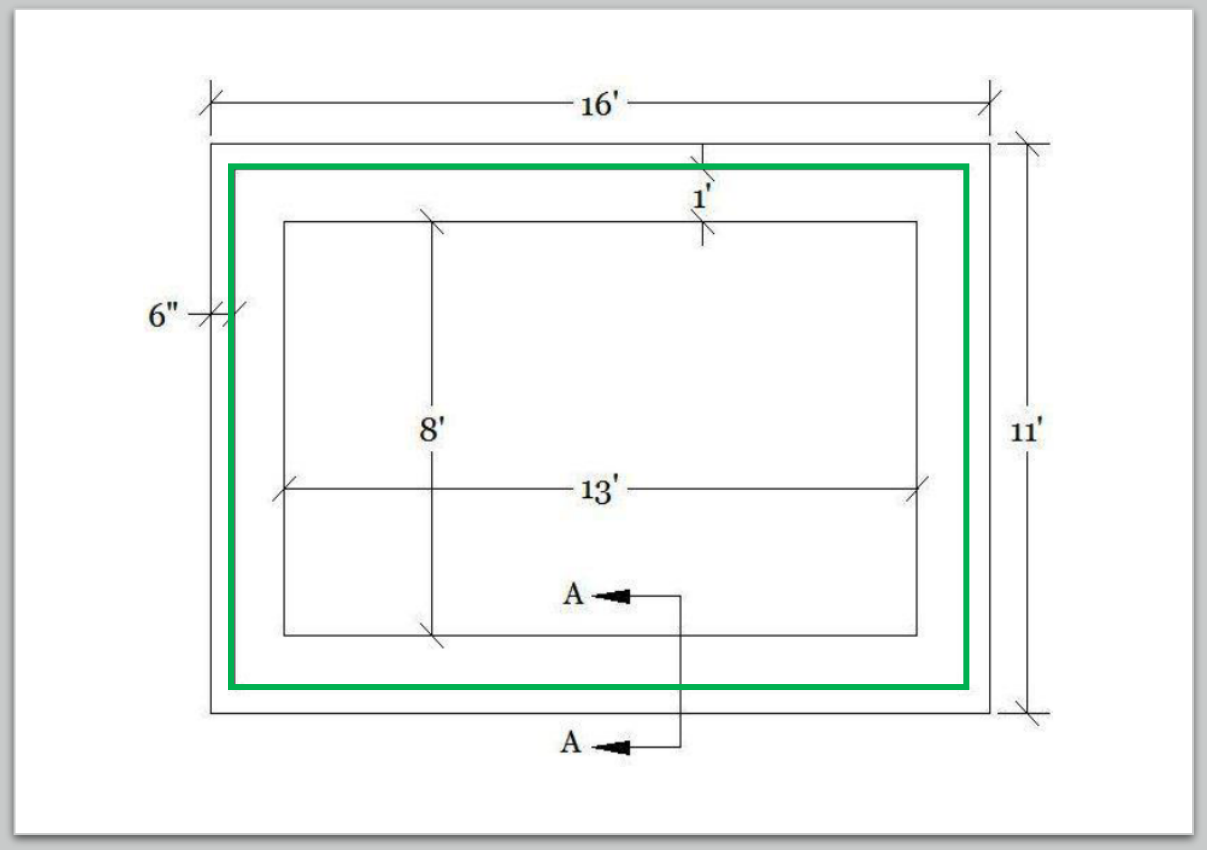
Cover Slab

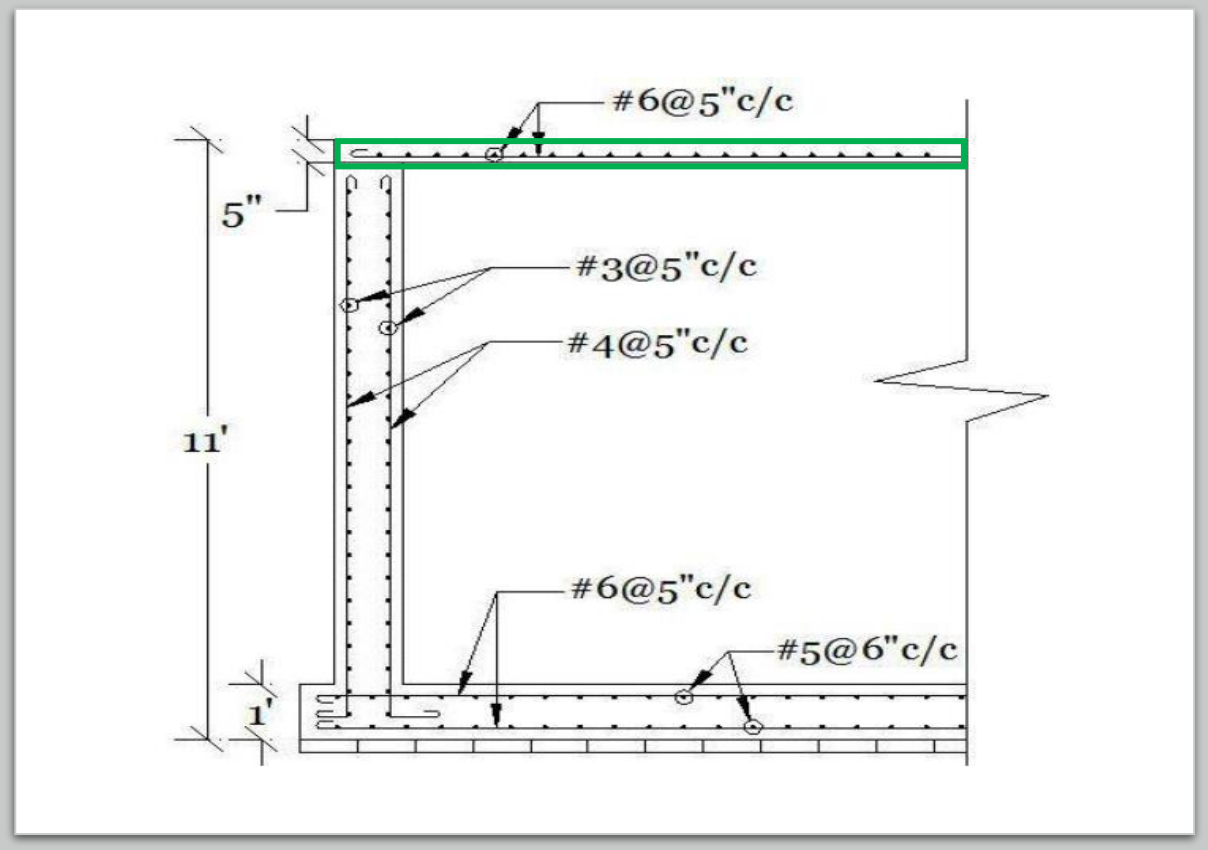
#6@5”c/c (bottom)

- No.= (15x12-2x3)/5+1= 36
- Length= (10x12-2x3+2x10.5x6/8)/12= 10.813ft.
- Total length= 389.268ft.

#6@5”c/c (top)

- No.= (10x12-2x3)/5+1= 24
- Length= (15x12-2x3+2x10.5x6/8)/12= 15.813ft.
- Total length= 379.512ft.





Bar Properties

Imperial Bar Size	"Soft" Metric Size	Mass per unit length		Nominal Diameter		Nominal Area	
		(lb/ft)	(kg/m)	(inch)	(mm)	($inch^2$)	(mm^2)
#2	#6	0.167	0.249	0.250	6.35	0.05	32
#3	#10	0.376	0.561	0.375	9.525	0.11	71
#4	#13	0.668	0.996	0.500	12.7	0.20	129
#5	#16	1.043	1.556	0.625	15.875	0.31	200
#6	#19	1.502	2.24	0.750	19.05	0.44	284

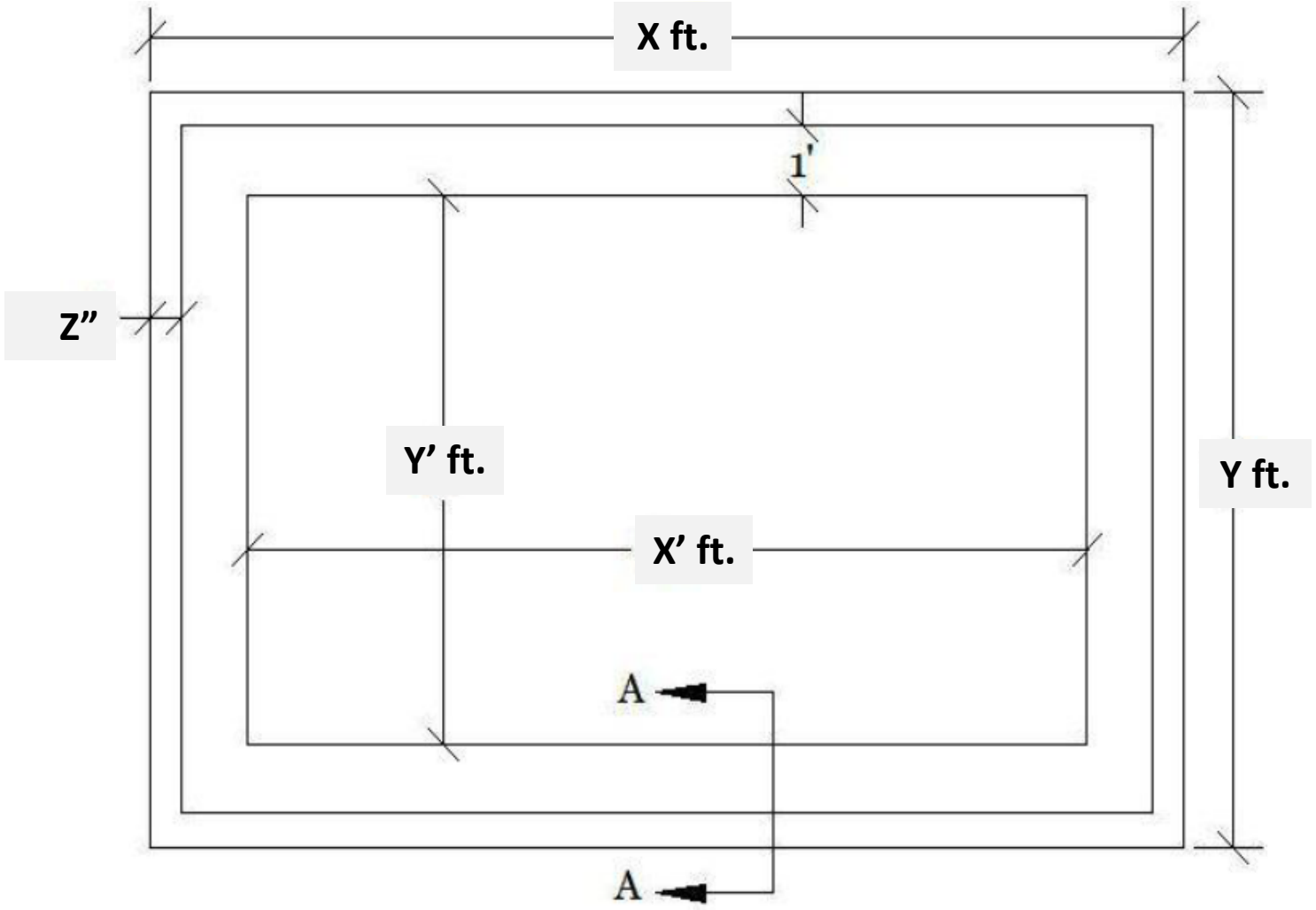
Summary

Bar	Length (ft)	Additional (2%)	Final Length (ft)	Weight/length (lb/ft)	Total weight (lb)
#3	2392	47.87	2440	0.376	918
#4	2611.54	52.23	2664	0.668	1780
#5	730.14	14.60	745	1.043	778
#6	1678.38	33.57	1712	1.502	2572



Assignment

Plan View



Section A-A

